Each person inherits two genes (one from each parent) that control the production of the ABO antigens. The genes for A or B antigens are dominant to the gene for O. The O gene is recessive, simply because it does not code for either the A or the B red blood cell antigens. The genes for A and B are often shown as I^A and I^B and the recessive gene for O is shown as the lowercase i. A person who is type A, therefore, may have inherited the A gene from each parent (I^A I^A), or the A gene from one parent and the O gene from the other parent (I^Ai). Likewise, a person who is type B may have the genotype I^BI^B or I ^{B}i. It follows that a type O person inherited the O gene from each parent , whereas a type AB person inherited the A gene from one parent and the B gene from the other (I^A I^B).

The immune system exhibits tolerance to its own red blood cell antigens. People who are type A, for example, do not produce anti-A antibodies. However, they do make antibodies against the B antigen and, conversely, people with blood type B make antibodies against the A antigen. This is believed to result from the fact that antibodies made in response to some common bacteria present in the gut (normal flora) which are similar to A or B antigens. People who are type A, therefore, acquire antibodies that can react with B antigens by exposure to these bacteria, but they do not develop antibodies that can react with A antigens because tolerance mechanisms prevent this. People who are type AB develop tolerance to both of these antigens, and thus do not produce either anti-A or anti-B antibodies. Those who are type O, by contrast, do not develop tolerance to either antigen; therefore, they have both anti- A and anti-B antibodies in their plasma.

Isoantibodies: Iso means belonging to the same species. Isoantibody is an antibody produced by one individual that reacts with the antigen of another individual of the same species. Antibody – A and Antibody– B are called Isoantibodies. Both of these anti-A and anti-B antibodies are of Ig M type. Rh antibody (Antibody - D) is of Ig G type.

Isoantigens: An antigen of an individual which is capable of eliciting an immune response in individuals of the same species who are genetically different and who do not possess that antigen is called isoantigen. It is also otherwise known as alloantigens. A-Antigen and B-Antigen are examples of isoantigens.

Natural antibodies: Humans form antibodies against the blood group antigens they do not express.These antibodies are called naturally occurring antibodies or isoagglutinins. Antibody production starts at 3 months of age, reaches its highest level during adult and decreases with advancing age.

Test Procedure

Basically, a sample of blood is mixed separately with anti-A antibodies, anti-B antibodies and Rh antibodies. If the red cells clump together with anti-A antibodies, then it indicates the presence of A antigens in the red blood cells and the person belongs to A group. Similarly, if agglutination reaction occurs with anti-B antibodies then it indicates the presence of B antigen.

When agglutination is found in both anti-A and anti-B antibodies it indicates that the person belongs to AB group. If no agglutination is found with both antibodies of A and B

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